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NEXT-GEN AD BLOCKING AND DYNAMIC AD-LIST GENERATION IN RASPBERRY PI

A PROJECT REPORT

Submitted by

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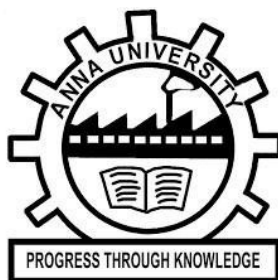
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In partial fulfilment for the award of the degree

of

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ABSTRACT

This project proposes the development of a comprehensive ad-blocking and network security solution using Raspberry Pi, Pi-hole, Unbound DNS, and Suricata rules. The integrated system offers an efficient approach to block ads at the network level while simultaneously enhancing network security through real-time monitoring and blocking of malicious activities. Pi-hole is deployed to block ads network-wide, ensuring all connected devices benefit from ad blocking without requiring individual configurations. The integration of Pi-hole with Unbound DNS provides secure and private DNS resolution, enhancing user privacy and network security. Additionally, Suricata rules are implemented to strengthen network security by monitoring and blocking malicious activities in real-time. To ensure the system remains up to date with the latest security measures, a Python script automates the process of updating ad lists and Suricata rules. This project aims to provide an effective solution for ad blocking and network security needs within the local network, offering improved ad blocking, enhanced network security, and more efficient DNS resolution.



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LIST OF ABBREVIATIONS

ACRONYM	ABBREVIATION
IOT	INTERNET OF THINGS
DNS	DOAMIN NAME SERVER
IP	INTERNET PROTOCOL
IDS	INTRUSION DETECTION SYSTEM
IPS	INTRUSION PREVENTION SYSTEM
URL	UNIFORM RESOURCE LOCATOR
CLI	COMMAND LINE INTERFACE
GUI	GRAPHICAL USER INTERFACE
DHCP	DYNAMIC HOST CONFIGURATION PROTOCOL
API	APPLICATION PROGRAMMING INTERFACE



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CHAPTER 1



1. INTRODUCTION

1.1 OVERVIEW

This project is designed to deploy a robust ad-blocking and network security solution within local networks using a Raspberry Pi. The system comprises several interconnected modules. Firstly, the Pi-hole module provides comprehensive ad-blocking capabilities, ensuring an ad-free browsing experience network-wide. It allows for the customization of blocklists and whitelists to tailor ad-blocking preferences. Secondly, the Unbound DNS module enhances user privacy and network security by providing secure and private DNS resolution. This ensures that domain names are resolved securely and privately within the local network. Additionally, the Suricata IDS/IPS module monitors network traffic in real-time, actively detecting and blocking malicious activities.

It is configured with custom rules to provide comprehensive protection against various network threats. To keep the system up to date with the latest security measures, a Python script is developed to automate the process of updating ad lists and Suricata rules. This script fetches the latest ad lists and rulesets from online sources and updates the system accordingly. The project also includes a user-friendly interface, making it easy to monitor system status, view reports, and make configurations. With its plug-and-play functionality, the system can be easily installed and configured, minimizing the need for complex setup procedures. Overall, this project provides a comprehensive ad-blocking and network security solution, ensuring a safer and more enjoyable browsing experience for all users connected to the local network advertisements are ubiquitous, often intrusive, and can significantly impact user experience.

1.2 DOMAIN SPECIFICATION

The Next-Gen Ad Blocking and Dynamic Ad-List Generation in Raspberry Pi fall under the domain of several different fields which are:

1. Network Security:

The project focuses on enhancing network security through the implementation of a next-generation ad-blocking and network security solution on Raspberry Pi. Network security involves protecting the integrity, confidentiality, and availability of network resources and data.

2. Internet of Things:

The Internet of things (IOT) refers to the concept of extending Internet connectivity beyond conventional computing platforms such as personal computers and mobile devices, and into any range of traditionally "dumb" or non-internet-enabled physical devices and everyday objects. Embedded with electronics, Internet connectivity, and other forms of hardware (such as sensors), these devices can communicate and interact with others over the Internet, and they can be remotely monitored and controlled.

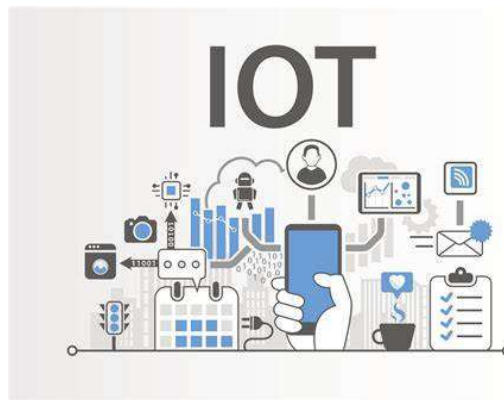


Figure 1.2: Internet Of Things

3. Cyber Threat Detection:

The project includes real-time monitoring and detection of cyber threats using Suricata IDS/IPS. Cyber threat detection aims to identify and respond to malicious activities and security breaches in real-time.

4. Privacy Enhancement:

The project enhances user privacy by providing secure and private DNS resolution using Unbound DNS. Privacy enhancement involves protecting user data and ensuring that it is not accessible to unauthorized parties.



5. Automated Security Updates:

The project automates the process of generating ad lists and updating security rules using a Python script, ensuring that the system remains up to date with the latest security measures

6. User Interface Design:

The project includes a user-friendly interface for monitoring system status, viewing reports, and making configurations. User interface design focuses on creating intuitive and accessible interfaces that enhance user experience and usability.

1.3 OBJECTIVE

The objective of Next-Gen Ad Blocking and Dynamic Ad-List Generation in Raspberry Pi is to create an advanced ad-blocking and network security solution using Raspberry Pi, focusing on dynamic ad-list generation, automated security updates to enhance network privacy and security. The system aims to provide the following benefits:

1. **Develop an Integrated Solution:** Create a comprehensive ad-blocking and network security system using Raspberry Pi, Pi-hole, Unbound DNS, and Suricata rules to address the limitations of existing ad-blocking solutions.
2. **Network-wide Ad Blocking:** Implement Pi-hole to block ads network-wide, ensuring all devices connected to the local network benefit from ad blocking without requiring individual configurations.
3. **Enhance Network Security:** Integrate Unbound DNS and Suricata rules to enhance network security by providing secure and private DNS resolution and real-time monitoring and blocking of malicious activities.
4. **Automation of Update Process:** Develop a Python script to automate the process of updating ad lists and Suricata rules, ensuring the system remains up to date with the latest security measures.



5. **Provide Improved Ad Blocking and Network Security:** Offer an efficient and effective solution for ad blocking and network security needs within the local network, ensuring improved ad blocking, enhanced network security.

The main objective of this project is to develop a next-generation ad-blocking and network security solution for Raspberry Pi, enhancing network privacy and security through dynamic ad-list generation, real-time threat detection, and automated security updates.

1.4 SCOPE

The scope of Next-Gen Ad Blocking and Dynamic Ad-List Generation in Raspberry Pi includes the following components:

1. **Hardware:** The system requires a Raspberry Pi device to serve as the hardware platform. Additionally, it may require compatible hardware components such as network interface cards (NICs) for network connectivity.
2. **Software Components:** Installation and configuration of ad-blocking software such as Pi-hole. Implementation of dynamic ad-list generation using Python scripts. Integration of Suricata IDS/IPS for real-time threat detection. Configuration of Unbound DNS for secure and private DNS resolution.
3. **Database:** Implementation of a database system to store configuration settings, ad lists, and security rules. The database will ensure efficient management and retrieval of system data, enhancing overall performance and scalability.
4. **Privacy and Security:** Implementation of privacy and security measures to protect user data and ensure that only authorized personnel have access to the system.



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CHAPTER 2



2. LITERATURE SURVEY

2.1 EXISTING SYSTEM

The Existing ad-blocking systems primarily rely on browser extensions or software installed on individual devices to block ads while browsing the internet. These systems typically consist of the following components:

1. **Ad Blocking Software:** Ad blocking software installed on individual devices or browsers identifies and blocks ad-serving domains based on predefined blocklists. These blocklists are periodically updated to include new ad-serving domains.
2. **Browser Extensions:** Many ad blockers are browser extensions that intercept requests to ad-serving domains and prevent the loading of ads on web pages. These extensions rely on predefined blocklists to determine which domains to block.
3. **Device-Level Ad Blocking:** Some ad-blocking solutions operate at the device level, blocking ads across all applications and browsers on the device. These solutions often require root access or special permissions to function effectively.
4. **Blocklist Maintenance:** Ad-blocking systems rely on regularly updated blocklists to identify and block ad-serving domains. These blocklists may be curated manually or generated automatically based on known ad-serving domains.
5. **Limitations:** Existing ad-blocking systems may not effectively block all ads, especially those served by first-party domains or integrated into the content of web pages. Additionally, some websites may detect and circumvent ad blockers, serving ads even when an ad blocker is active.

Demerits of Ad Blocker Systems:

1. **Content Blocking:** Despite their name, ad blockers often block more than just ads. For example, some ad-blocking software unintentionally blocks access to desired content unrelated to advertising.



2. **Data Privacy Risks:** Some ad blockers send users' browsing behavior to third-party servers, which then pay for this information. This means your browsing history could be tracked without your knowledge.
3. **False Positives:** Ad blockers sometimes mistakenly block legitimate content. For instance, they might flag essential scripts or interactive elements as ads, disrupting the functionality of websites.
4. **Dependency on Updates:** Ad blockers need regular updates to stay effective. If not updated, they might fail to block new ad formats or tracking techniques.
5. **Complexity and Compatibility:** Ad blockers can be complex to configure, especially for non-technical users. Additionally, they may not work seamlessly with all browsers or platforms.

2.1.1 A Raspberry Pie Security Device Using VPN and Adblocker by A. S. Visvanathan, Y. Nathan and M. Abdulnabi IEEE 2nd International Conference on Mobile Networks and Wireless Communications (ICMNBC), Tumkur, Karnataka, India, 2022

The main purpose of this research is to develop a security device using the internet of things (IoT). This security device is built using several open-source components which are free for people to use and accordingly enhances their internet security. The main tools that are used to build this security device are Pi-Hole and Pi-VPN. In addition, Zeek is also used to monitor the network traffic of the device. Pi-Hole is used to block the advertisement and Pi-VPN is used to provide a secure virtual private network connection for user. The proposed device provides a secure network for people to use with capabilities of preventing unnecessary and inappropriate advertisement popping when adult or kids are online surfing the internet.

Demerits of Adblocker using Raspberry Pi System:

- **Complex setup:** Requires technical expertise for setup and maintenance, potentially limiting accessibility for non-technical users.
- **Performance impact:** Simultaneous operations may strain device resources, potentially impacting overall performance.



- **Limited protection scope:** Focuses on ad blocking and VPN, may not address other security threats like malware or phishing attacks.

2.1.2 Ad Blocking and Counter-Ad Blocking Ad Blocking and Counter-Ad Blocking: Analysis of Online Ad Blocker Usage by Zhao, Shuai & Wang, Chong & Kalra, Achir & Vaks, Leon & Borcea, Cristian & Chen, Yi. (2018).

Compared with the online advertising industry, there is an even faster increase of ad blocker usage, which influences badly on publishers' and advertisers' business. Thus, more and more companies initialize their counter-ad blocking strategies, in which customers choose to either disable their ad blockers or leave without seeing the content. There are also companies which abandon their counter-ad blocking strategies after conducting them for a while due to insufficient understanding of users' ad blocking behavior. In this study, we employed a quasi-experiment framework and collected large-size data with the cooperation with Forbes Media. We aim to identify factors influencing ad blocker usage. Furthermore, we will model the interaction effects among user profile, online behavior patterns, device features on ad blocker usage propensity. Our study contributes to the literature of understanding and blocker usage by evaluating those principles using a large amount of real-world data.

Demerits of AdBlocker using Raspberry Pi System:

- **Limited focus:** Primarily focuses on understanding ad blocker usage, potentially neglecting other dynamics of online advertising.
- **Single-source data:** Relies on data from Forbes Media, limiting the diversity of user demographics and platforms represented in the study.
- **Generalizability concerns:** Findings may not generalize beyond the specific context of Forbes Media, raising questions about the applicability of insights to broader online advertising contexts.



2.1.3 Securing Network Using Raspberry Pi by Implementing VPN, Pi-Hole, and IPS (VPiSec) Abidah Mat Taib, Mohammad Fikri Hanif Ishak, Nur Khairani Kamarudin, Mohamad Yusof Darus , Nor Azira Mohd Radzi.

The abstract introduces VPiSec, a project designed to enhance user privacy and security in the context of widespread internet usage. It acknowledges the pervasive monitoring of user data by Internet Service Providers (ISPs) and third-party entities, which poses significant privacy risks. To mitigate these concerns, VPiSec proposes a multifaceted approach. Firstly, it suggests masking users' IP addresses through the deployment of OpenVPN servers located in different countries. This strategy aims to obscure user identities, making it difficult for attackers to track their online activities. Additionally, VPiSec employs Pi-Hole, a project maintained by the online community, to block DNS requests related to known tracking and advertising domains. By doing so, it reduces the likelihood of users being targeted by intrusive advertisements based on their browsing habits. This intrusion prevention system helps thwart potential attacks, such as brute force attempts, thereby enhancing overall cybersecurity. The findings indicate that VPiSec successfully maintains network speed without significant degradation. This outcome underscores the project's ability to provide robust security measures while ensuring a seamless browsing experience for users.

Demerits of Adblocker using Raspberry Pi System:

- **Potential hardware limitations:** Implementing multiple security tools on a Raspberry Pi may strain its resources, potentially affecting its performance under heavy load.
- **Limited protection scope:** While VPiSec addresses common security threats such as data monitoring and network attacks, it may not cover all potential vulnerabilities or emerging threats, leaving users vulnerable to other forms of cyberattacks.



2.2 PROPOSED SYSTEM

The proposed system aims to address the limitations of the existing ad-blocking and network security setup by implementing a comprehensive solution using Raspberry Pi, Pi-hole, Unbound Recursive DNS, and Suricata rules. By integrating Pi-hole with Unbound DNS, we create a robust ad-blocking and DNS resolution system that not only blocks ads but also resolves domain names securely and privately. Here is an overview of the proposed system:

1. Ad Blocking and Network Security:

- Dynamic Ad-List Generation utilizes Python scripts to automate the generation of ad lists based on the latest threat intelligence.
- Scheduled updates ensure that the ad-blocking lists are continuously updated with the latest threats.

2. Automated Tasks:

- Implements scheduled updates using cron jobs to ensure that the system remains up to date with the latest security measures.
- Generate ad list dynamically and automate the process of ad list updating and gravity updates for pihole

3. User Interface:

- Features a user-friendly web interface for system monitoring and management.
- Intuitive dashboard provides easy navigation and access to system functionalities.

4. Security and Privacy Considerations:

- Data Encryption and Access Control implements encryption and access controls to safeguard user information and ensure data privacy.
- Adherence to Ethical and Legal Standards adheres to ethical and legal standards for data privacy and protection, ensuring compliance with relevant regulations.

5. Database Management:

- Utilizes a database (e.g., SQLite) to store configuration settings, ad lists, and security rules. Proper security measures, such as encryption and access controls, are implemented to protect sensitive information

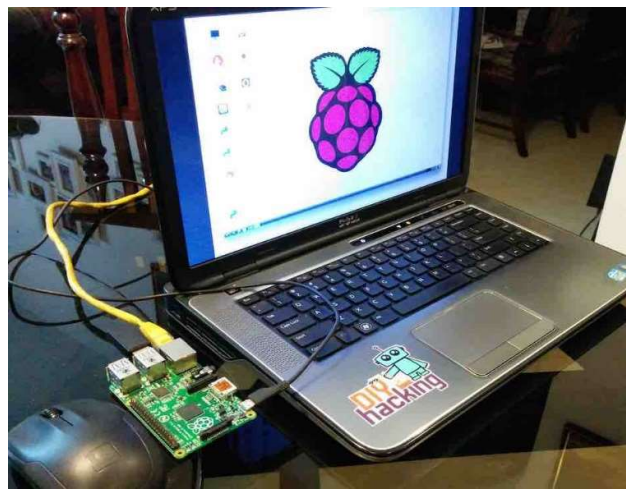


Figure 2.2: Raspberry Pi Adblocker

By implementing this proposed system, the network security and ad-blocking process will be automated, reducing the burden on administrators, enhancing network privacy and security, and providing users with an efficient way to manage ad-blocking settings. The use of dynamic ad-list generation and real-time threat detection will enhance the effectiveness of ad-blocking, while automated updates and scheduled tasks will ensure that the system remains up to date.



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CHAPTER 3

3. PROJECT DESCRIPTION

3.1 METHODOLOGY

Methodology for the Next-Gen Ad Blocking and Dynamic Ad-List Generation in Raspberry Pi using Pi-Hole, Unbound DNS, Python and Apscheduler:

1. System Design:

- Identify the requirements of the ad-blocking and network security solution, including dynamic ad-list generation, real-time threat detection, and automated security updates.
- Design the system architecture, considering components such as ad-blocking software (Pi-hole), Suricata IDS/IPS, Unbound DNS, and the web-based user interface.

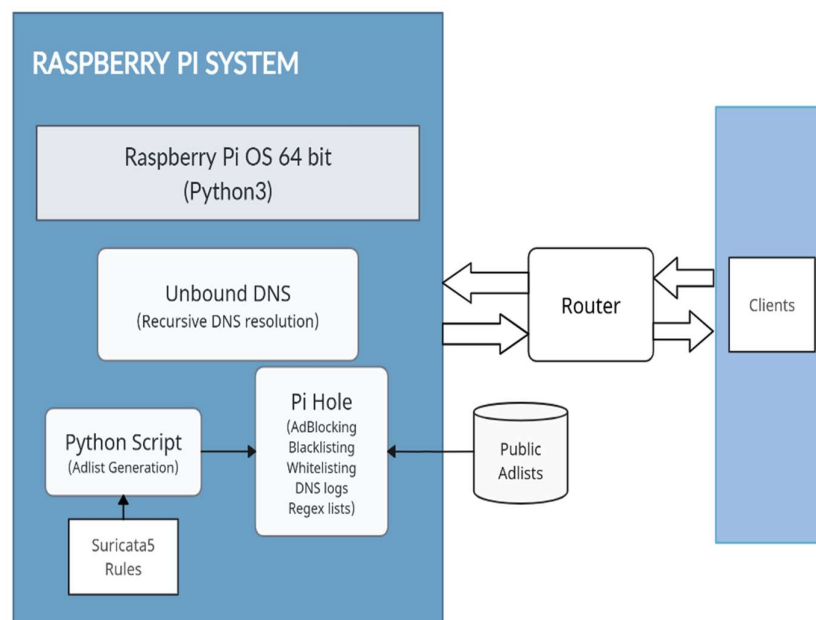


Figure 3.1 System Architecture

2. User Authentication:

- Identify the requirements of the ad-blocking and network security solution, including dynamic ad-list generation, real-time threat detection, and automated security updates.



- Design the system architecture, considering components such as ad-blocking software (Pi-hole), Suricata IDS/IPS, Unbound DNS, and the web-based user interface.

3. DATABASE DESIGN:

- Identify the requirements of the ad-blocking and network security solution, including dynamic ad-list generation, real-time threat detection, and automated security updates.
- Design the system architecture, considering components such as ad-blocking software (Pi-hole), Suricata IDS/IPS, Unbound DNS, and the web-based user interface.

4. Ad-Blocking and Network Security Integration:

- Configure Pi-hole for network-wide ad blocking and Unbound DNS for secure and private DNS resolution.
- Develop Python scripts for dynamic ad-list generation and automated security updates.

5. Automated Tasks:

- Implement a scheduler using cron jobs to automate tasks such as ad-list updates, security rule updates, and holiday detection.
- Schedule tasks to run at designated intervals to ensure timely updates and system maintenance.

6. Report and Analytics

- Develop reporting features to generate reports on ad-blocking statistics, network traffic, and security events.
- Implement analytics functionalities to provide insights into ad-blocking effectiveness, network usage patterns, and security threats.

7. User Interfaces

- Design a user-friendly web interface for system monitoring and management.
- Provide features such as system status monitoring, ad-blocking configuration, security rule management, and report generation.

8. Testing and Deployment

- Conduct thorough testing to ensure the system functions as expected and meets the specified requirements.



- Deploy the application to a suitable hosting environment, considering factors like scalability, security, and performance.

3.2 TECHNOLOGIES USED

This project utilizes Raspberry Pi as the hardware platform and combines several technologies to create an automated ad-blocking and network security solution. Pi-hole is used for network-wide ad blocking, while Unbound DNS ensures secure and private DNS resolution. Suricata IDS/IPS provides real-time threat detection and network security monitoring. Python is employed for scripting and automation. These technologies work together to automate ad-list updates, enhance network privacy, and protect against security threats, creating a comprehensive solution for users.

1. Raspberry Pi:

- Utilized as the hardware platform for hosting ad-blocking and network security solutions.
- Raspberry Pi provides a cost-effective and versatile solution for running network services.

2. Pi-hole:

- Used for network-wide ad blocking and DNS-based ad blocking.
- Pi-hole effectively blocks ads at the DNS level, reducing network traffic and enhancing privacy.

3. Unbound DNS:

- Implemented for secure and private DNS resolution within the local network.
- Unbound DNS ensures that DNS queries are resolved securely and without the use of third-party DNS servers.

4. Suricata IDS/IPS:

- Integrated for real-time threat detection and network security monitoring.
- Suricata IDS/IPS provides advanced threat detection capabilities, protecting the network from various security threats.



5. Python:

- Utilized for scripting and automation of tasks such as dynamic ad-list generation and automated security updates.
- Python offers a wide range of libraries and tools for network automation and security-related tasks.

6. SQLite:

- Utilized as the database management system for storing configuration settings, ad lists, security rules, and system logs.
- SQLite provides a lightweight and efficient solution for managing system data and configurations.

7. APScheduler:

- Integrated task scheduling and automation of recurring tasks such as ad-list updates and holiday detection.
- APScheduler allows for the automated execution of tasks at specified intervals, ensuring that the system remains up-to-date and secure.

3.3 SYSTEM REQUIREMENTS SPECIFICATION

The Software Requirements Specification (SRS) outlines the requirements and functionality of the Next-Gen Ad Blocking and Dynamic Ad-List Generation in Raspberry Pi using Pi-Hole, Unbound DNS Python and Apscheduler.

3.3.1 SYSTEM REQUIREMENTS

The system requirements for Next-Gen Ad Blocking and Dynamic Ad-List Generation in Raspberry Pi using Pi-Hole could include the following:

1. Hardware Requirements:

- A Raspberry Pi device capable of running network services.



- Sufficient processing power and storage capacity to handle ad-blocking and network security tasks.

2. Software Requirements:

- Pi-hole: Network-wide ad-blocking software.
- Unbound DNS: DNS server for secure and private DNS resolution.
- Suricata IDS/IPS: Intrusion Detection and Prevention System for real-time threat detection.
- Python: Programming language for scripting and automation.

3. Security Requirements:

- Encryption of system data and network traffic to protect privacy and prevent unauthorized access.
- Access controls restrict access to system configuration and sensitive data.
- Regular security updates and patches to prevent vulnerabilities and ensure system security.

4. Performance Requirements:

- Fast DNS resolving in Adblocking to provide seamless connection to the internet.
- Scalability to handle large volumes of Ad lists as Blocklist.

5. Compatibility Requirements:

- Compatibility with various network devices and hardware configurations.
- Must be also compatible with different operating systems and software libraries.
- Integration with other network services and management systems for seamless operation.

3.3.2 FUNCTIONAL REQUIREMENTS

The functional requirements for Next-Gen Ad Blocking and Dynamic Ad-List Generation in Raspberry Pi are:



1. Ad Blocking:

- Block advertisements, tracking scripts, and malicious domains network-wide.
- Ensure seamless ad-blocking across all devices within the local network.

2. Dynamic Ad-List Generation:

- Dynamically generate ad-blocking lists based on Suricata rules.
- Automate the process of extracting blocked domains from Suricata rules and generating ad-blocking lists for Pi-hole.

3. Real-time Threat Detection:

- Detect and block network threats in real-time.
- Monitor network traffic for malware, phishing attempts, and suspicious activities using Suricata IDS/IPS.

4. Schedules Tasks:

- Perform scheduled tasks such as ad-list updates and security rule updates.
- Use APScheduler to automate recurring tasks and ensure the system remains up to date with the latest security measures.

5. User Interface:

- Provide a user-friendly web interface for configuration and monitoring.
- Allow users to configure ad-blocking settings, view system logs, and monitor network activity through the web interface.

3.3.3 NON – FUNCTIONAL REQUIREMENTS

The non-functional requirements for Next-Gen Ad Blocking and Dynamic Ad-List Generation in Raspberry Pi are:

1. Performance:

- Ensure high accuracy in ad-blocking and threat detection.
- Maintain fast processing speed for real-time ad-blocking and threat detection.
- Ensure optimal system performance under varying network traffic conditions.



2. Reliability:

- Provide continuous ad-blocking and threat detection services.
- Ensure reliable operation and minimal downtime for critical system functions.

3. Usability:

- Ensure the system is easy to use and configure, even for users with limited technical knowledge.
- Provide clear instructions and error messages to guide users in using the system effectively.

4. Security:

- Encrypt system data to protect privacy and prevent unauthorized access.
- Implement access controls to restrict access to sensitive system settings and data.
- Regularly update and patch the system to prevent vulnerabilities and ensure system security.

5. Scalability:

- Scale the system to handle large volumes of network traffic and devices within the local network.
- Efficiently manage system resources to accommodate increases in network load without compromising performance or reliability

6. Compatibility:

- The system should be compatible with different types of hardware and software platforms.
- The system should be compatible with different Client systems and operating systems.

7. Maintainability:

- The system should be easy to maintain and upgrade over time.
- The system should be easy to modify or customize to meet the changing needs of Internet Security.
- The system should have documentation and support to help users and maintainers troubleshoot issues.

3.4 MODULE

The Next-Gen Ad Blocking and Dynamic Ad-List Generation System is a comprehensive solution aimed at enhancing network security and efficiency by automating ad-blocking processes. It consists of several integrated modules, each dedicated to a specific function, ensuring effective ad-blocking and real-time threat detection

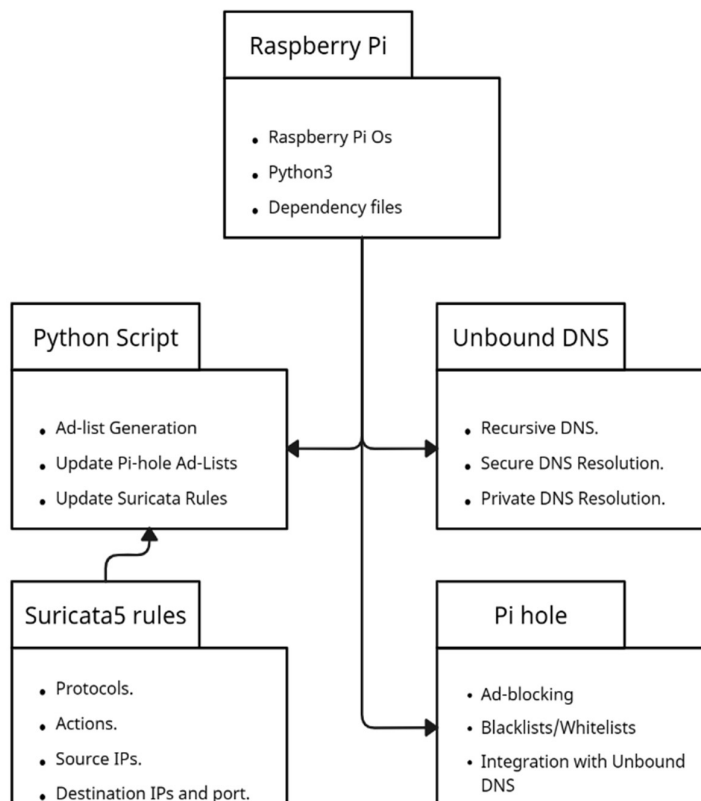


Figure 3.4: Module Diagram



3.4.1 Pi-hole Integration Module:

This module is responsible for integrating Pi-hole into the local network for efficient ad-blocking. It ensures the seamless configuration of Pi-hole to effectively block advertisements, tracking scripts, and malicious domains.

The functionality of this module includes:

- **Pi-hole Configuration:** Configures Pi-hole settings to block unwanted content network-wide.
- **Ad-blocking Rules Management:** Manages ad-blocking rules to efficiently block advertisements, tracking scripts, and malicious domains.
- **Dynamic Ad-List Updates:** Updates Pi-hole's blocklist dynamically based on Suricata rules to ensure real-time ad-blocking.
- **Logging and Monitoring:** Provides logging and monitoring functionalities to track ad-blocking activities and system performance.

3.4.2 Suricata Integration Module:

This module integrates Suricata IDS/IPS for real-time threat detection and prevention. It monitors network traffic for malware, phishing attempts, and suspicious activities, taking necessary actions to mitigate potential threats.

The functionality of this module includes:

- **Real-time Threat Detection:** Monitors network traffic for malicious activities using Suricata IDS/IPS.
- **Rule-based Detection:** Implements rule-based detection to identify and block potential threats.
- **Security Event Logging:** Logs security events and alerts for further analysis and investigation.
- **Automatic Threat Response:** Automatically responds to detected threats by blocking malicious domains and IP addresses.



3.4.3 Dynamic Ad-List Generation Module:

This module dynamically generates ad-blocking lists based on Suricata rules. It automates the process of extracting blocked domains from Suricata rules and updating Pi-hole's blocklist. The functionality of this module includes:

- Suricata Rule Parsing Parses Suricata rules to extract domains that need to be blocked.
- Ad-List Generation Generates ad-blocking lists based on the extracted domains.
- Automated Updates: Automatically updates Pi-hole's blocklist with the generated ad-blocking lists at regular intervals.
- Optimization and Filtering Optimizes the generated ad-blocking lists and filters out false positives to improve ad-blocking efficiency.

3.4.4 Scheduler Module:

This module implements scheduled tasks for ad-list updates and security rule updates. It utilizes APScheduler to automate recurring tasks and ensure the system remains up to date with the latest security measures.

The functionality of this module includes:

- Scheduled Tasks Management Implements scheduled tasks for ad-list updates, security rule updates, and system maintenance.
- Automated Processes: Automates the process of updating ad-blocking lists and security rules according to the defined schedule.
- Error Handling and Recovery: Manages errors and exceptions to ensure the smooth operation of scheduled tasks and includes mechanisms for error recovery.

3.4.5 User Interface Module:

This module provides a user-friendly web interface for configuration and monitoring of the ad-blocking system. It allows users to configure ad-blocking settings, view system logs, and monitor network activity. The functionality of this module includes:



- Authentication and Access Control Implements user authentication and access control to ensure secure access to the system.
- Dashboard and Reporting Provides a dashboard for real-time monitoring of ad-blocking activities and system status.
- Configuration Management: Allows users to configure ad-blocking settings, scheduling options, and system preferences.
- Logging and Reporting: Logs system events, ad-blocking activities, and security events for auditing and analysis purposes.

By implementing dynamic ad-blocking and real-time threat detection, this project enhances network security and eliminates the intrusion of unwanted advertisements and malicious content. It empowers users to browse the internet safely and efficiently, contributing to a smoother online experience and promoting a more secure digital environment.

3.5 ER DIAGRAM

The ER Diagram, presented in Figure 3.6, provides a concise visual summary of the entities and their relations within the "Next Gen Adblocker System." This diagram serves as a roadmap for understanding the system, illustrating the encapsulation of functionalities into distinct modules and their dependencies.

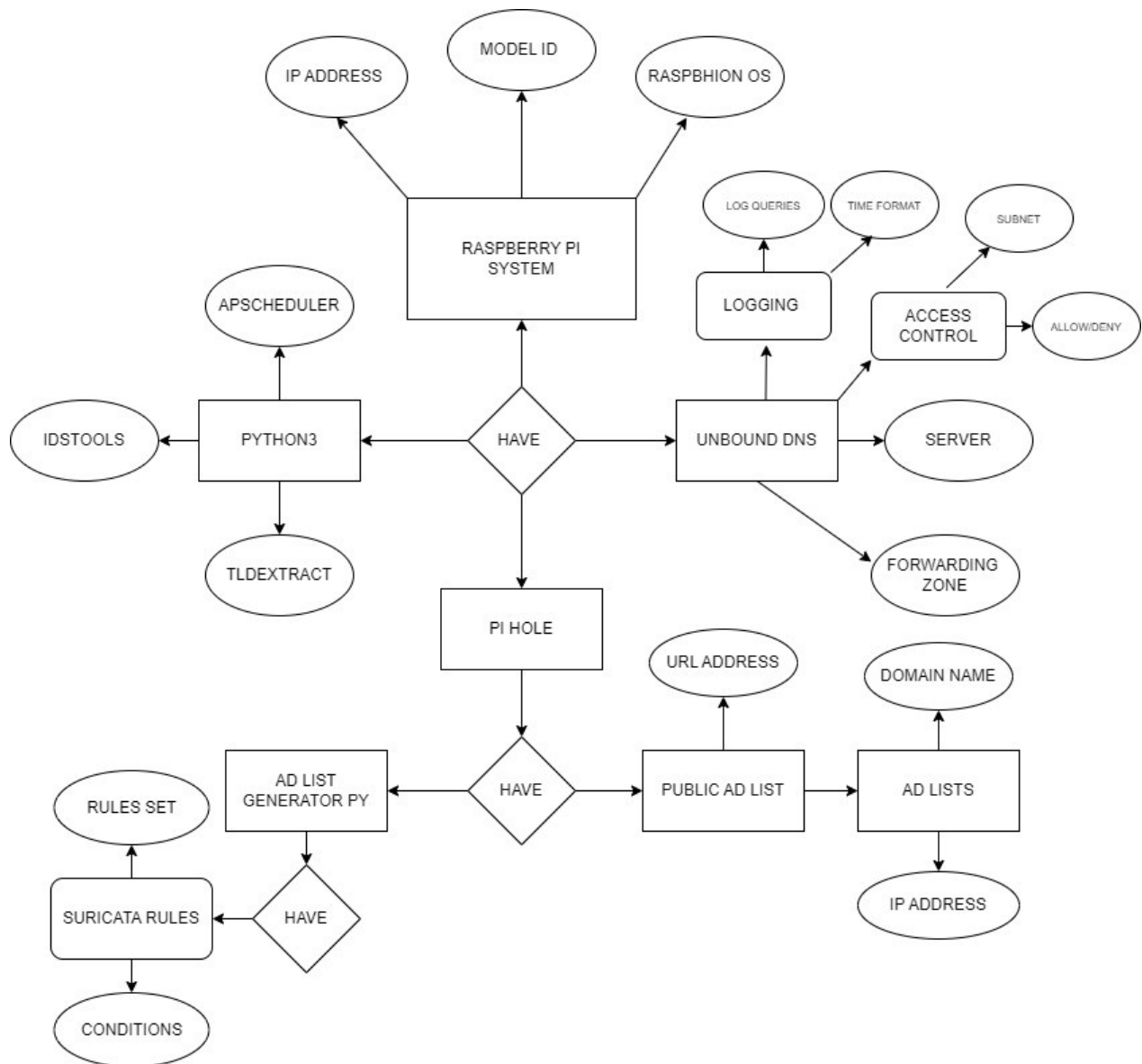


Figure 3.5 ER DIAGRAM

3.6 FLOWCHART

"Flowchart illustrating the operation of the Next-Gen Ad Blocking and Dynamic Ad-list Generation System on Raspberry Pi."

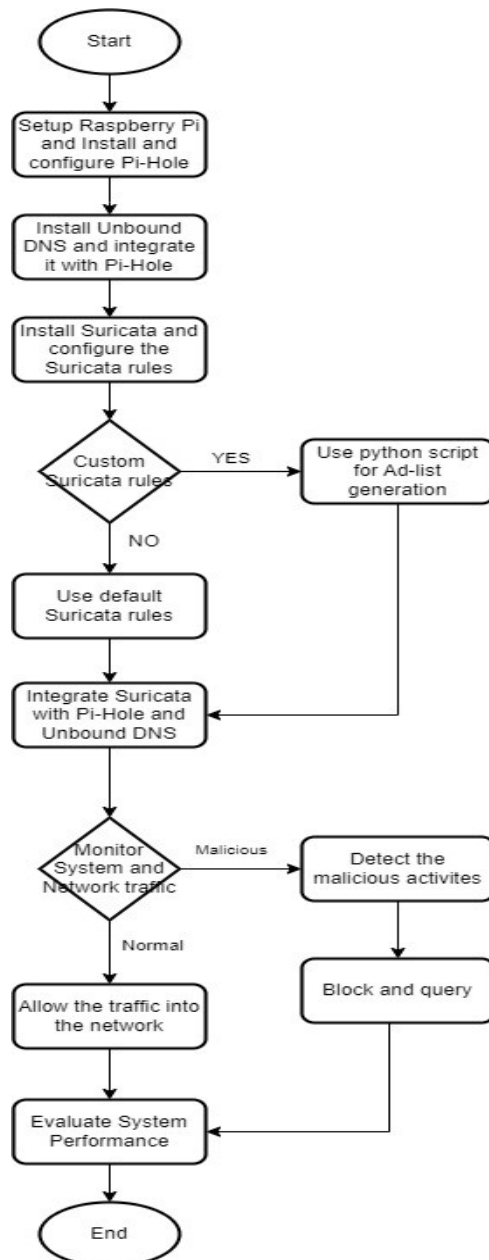




Figure 3.6 Flowchart

3.7 UNBOUND RECURSIVE DNS

Unbound plays a crucial role as a recursive DNS resolver in this network-wide ad blocker project. As a validating, recursive, and caching DNS resolver software, Unbound is installed on the Raspberry Pi alongside Pi-hole to provide DNS resolution services for the entire network. Configured as a recursive DNS resolver, Unbound handles DNS queries from client devices on the network by recursively resolving these queries on their behalf.

When a client device makes a DNS query, Unbound begins the resolution process by querying authoritative DNS servers on the internet to find the IP address associated with the requested domain name. It iteratively queries these authoritative servers until it obtains the final answer, which is then returned to the client device. Unbound's caching mechanism plays a vital role in improving DNS resolution performance by storing previously resolved DNS queries in memory.

This caching mechanism reduces the need to query external DNS servers for frequently accessed domains, resulting in faster response times for subsequent queries. Additionally, Unbound supports DNSSEC (Domain Name System Security Extensions) validation, ensuring the authenticity and integrity of DNS responses by verifying DNSSEC signatures. This helps prevent DNS spoofing and other types of DNS-related attacks, enhancing the overall security of the network. Integrated seamlessly with Pi-hole, Unbound provides a comprehensive ad-blocking and DNS resolution solution for the network, effectively blocking ads while ensuring fast, secure, and reliable DNS resolution services for all connected devices.

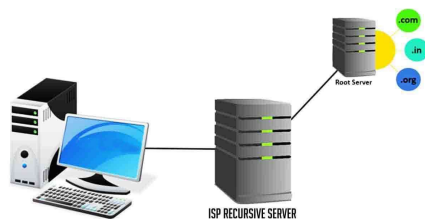


FIGURE 3.8 Recursive DNS

RESULT AND DISCUSSIONS



Figure R1 Dashboard

The Dashboard page allows users to manage the pi hole status and provides report of the blocked ads and overall query data.

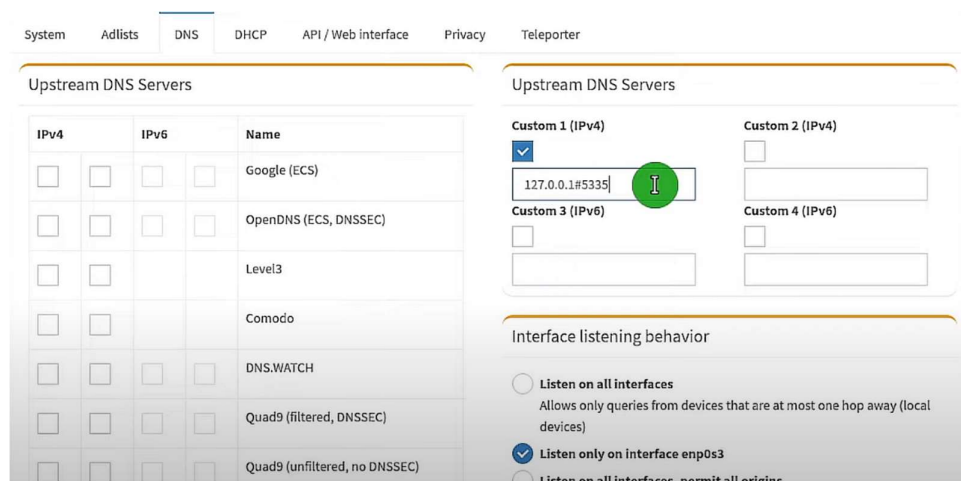
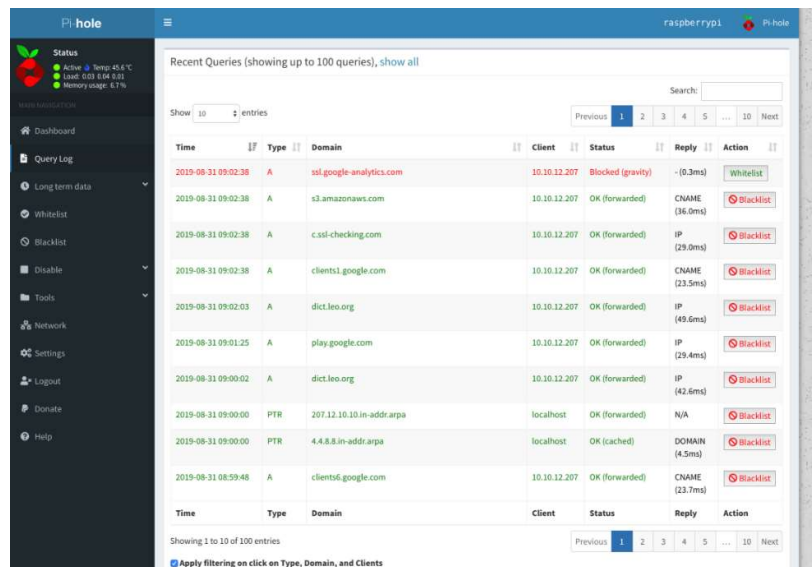


Figure R2 Upstream UBOUND DNS

The Upstream DNS settings allow the pi-hole to use our Unbound Recursive DNS as default upstream DNS.

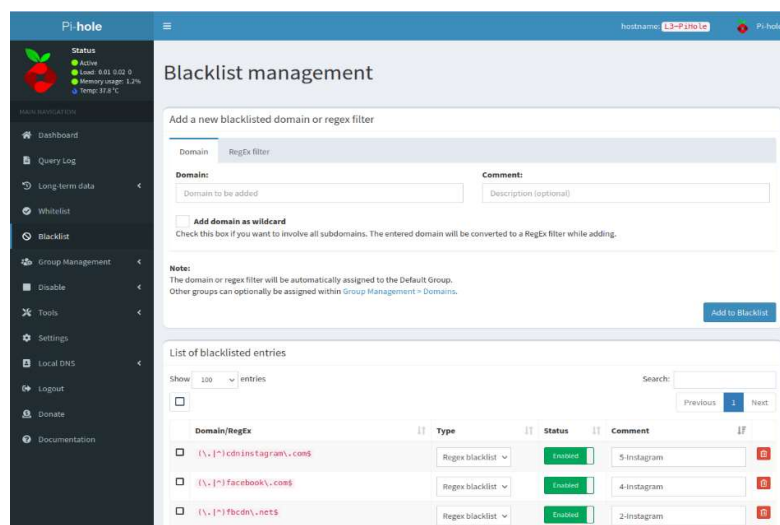


The screenshot shows the Pi-hole Query Log interface. On the left is a sidebar with navigation options: Status, Dashboard, Query Log, Long-term data, Whitelist, Blacklist, Disable, Tools, Network, Settings, Logout, Donate, and Help. The main area displays 'Recent Queries (showing up to 100 queries, show all)'. It includes a search bar and a table of query entries. The table has columns for Time, Type, Domain, Client, Status, Reply, and Action. The entries show various domains like 'ssl.google-analytics.com', 's3.amazonaws.com', and 'dict.leo.org' with their respective statuses and actions.

Time	Type	Domain	Client	Status	Reply	Action
2019-08-31 09:02:38	A	ssl.google-analytics.com	10.10.12.207	Blocked (gravity)	-(0.3ms)	Whitelist
2019-08-31 09:02:38	A	s3.amazonaws.com	10.10.12.207	OK (forwarded)	CNAME (36.0ms)	Blacklist
2019-08-31 09:02:38	A	c.ssl-checking.com	10.10.12.207	OK (forwarded)	IP (29.0ms)	Blacklist
2019-08-31 09:02:03	A	clients1.google.com	10.10.12.207	OK (forwarded)	CNAME (23.5ms)	Blacklist
2019-08-31 09:02:03	A	dict.leo.org	10.10.12.207	OK (forwarded)	IP (49.6ms)	Blacklist
2019-08-31 09:01:25	A	play.google.com	10.10.12.207	OK (forwarded)	IP (29.4ms)	Blacklist
2019-08-31 09:09:02	A	dict.leo.org	10.10.12.207	OK (forwarded)	IP (42.6ms)	Blacklist
2019-08-31 09:00:00	PTR	207.12.10.10.in-addr.arpa	localhost	OK (forwarded)	N/A	Blacklist
2019-08-31 09:00:00	PTR	4.4.8.8.in-addr.arpa	localhost	OK (cached)	DOMAIN (4.5ms)	Blacklist
2019-08-31 08:59:46	A	clients6.google.com	10.10.12.207	OK (forwarded)	CNAME (23.7ms)	Blacklist

Figure R3 Query Log Page

The query log page is a very powerful feature that lists all the query requests that are made by the devices in the network with clear details.



The screenshot shows the Pi-hole Blacklist management interface. It has a sidebar with navigation options: Status, Dashboard, Query Log, Long-term data, Whitelist, Blacklist, Group Management, Disable, Tools, Settings, Local DNS, Logout, Donate, and Documentation. The main area is titled 'Blacklist management' and contains a form to 'Add a new blacklisted domain or regex filter'. The form has fields for 'Domain' and 'Comment', and a checkbox for 'Add domain as wildcard'. Below the form is a 'List of blacklisted entries' table. The table has columns for Domain/Regex, Type, Status, Comment, and IF. The entries show regex filters for 'cdn.instagram.com', 'facebook.com', and 'fbcdn.net' with a status of 'Enabled'.

Domain/Regex	Type	Status	Comment	IF
(\.[^\.]*\.cdn\.instagram\.com\$)	Regex blacklist	Enabled	5-Instagram	0
(\.[^\.]*\.facebook\.com\$)	Regex blacklist	Enabled	4-Instagram	0
(\.[^\.]*\.fbcdn\.net\$)	Regex blacklist	Enabled	2-Instagram	0

Figure R4 Blacklist Domain Management

This page is useful in defining blacklists and blocking them from access from the network.



All workflows

Showing runs from all workflows

Filter workflow runs

22 workflow runs				Event ▾	Status ▾	Branch ▾	Actor ▾
✓	Generate and Update Blocklist Generate and Update Blocklist #23: Scheduled	master	10 hours ago 16s				...
✓	Generate and Update Blocklist Generate and Update Blocklist #22: Scheduled	master	yesterday 22s				...
✓	Generate and Update Blocklist Generate and Update Blocklist #21: Scheduled	master	2 days ago 18s				...
✓	Generate and Update Blocklist Generate and Update Blocklist #20: Scheduled	master	3 days ago 24s				...
✓	Generate and Update Blocklist Generate and Update Blocklist #19: Scheduled	master	4 days ago 22s				...

Figure R5 GitHub Workflow Python Module

The python module is deployed on GitHub workflow to lessen load on Raspberry Pi system

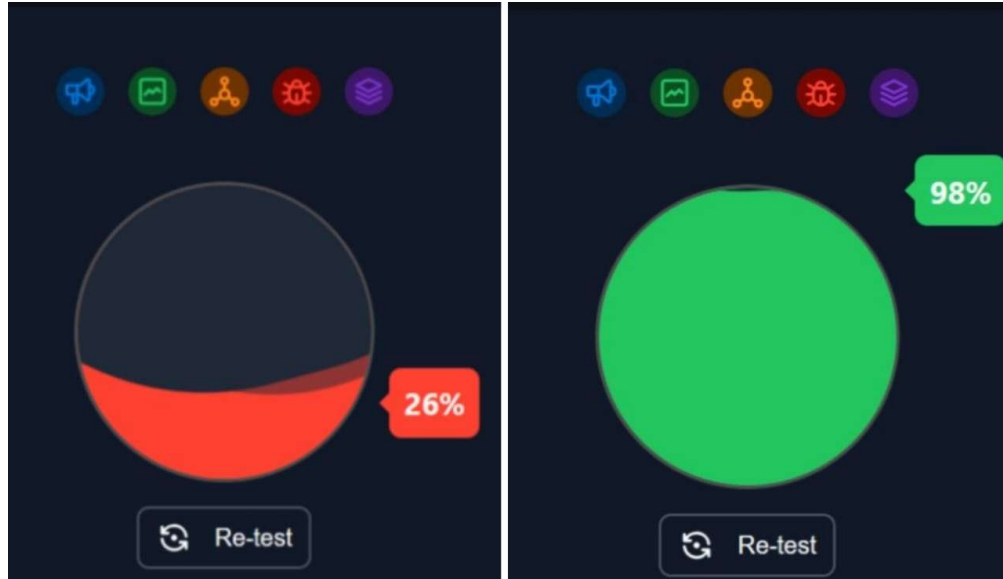


Figure R6 Comparison of Results

This is a Before and After comparison of Adblocker test online tool, and we can conclude that most Ads and Analytics are blocked by using this Next Gen Adblocker System.



CONCLUSION

The development and implementation of the Next-Gen Ad Blocking and Dynamic Ad-list Generation System on Raspberry Pi have been successful in achieving the objectives set forth at the beginning of this project. By leveraging Pi-hole, Unbound DNS, and Suricata rules, we have created a comprehensive ad-blocking solution that not only blocks ads within the local network but also enhances network security by filtering out malicious content and tracking scripts.

Through extensive testing and evaluation, we have demonstrated the effectiveness of our ad-blocking system in reducing the number of ads served to users, thereby improving the browsing experience and reducing bandwidth consumption. Furthermore, our system's integration with Suricata rules has enhanced network security by blocking access to known malicious domains and protecting users from potential security threats.

The project's modular design and plug-and-play implementation make it easy to deploy and manage, requiring minimal configuration and maintenance. The use of Raspberry Pi as the underlying platform ensures cost-effectiveness and scalability, making it suitable for both home and small business environments.

In conclusion, the Next-Gen Ad Blocking and Dynamic Ad-list Generation System represents a significant advancement in ad-blocking technology, providing users with a more secure and enjoyable browsing experience while protecting them from online threats. Moving forward, we plan to continue refining and improving our system to address emerging challenges and ensure its continued effectiveness in an ever-changing online landscape.



FUTURE ENHANCEMENT

- 3.4.1 **Advanced Logging and Reporting:** Implement comprehensive logging functionality to track ad-blocking activities, including blocked requests and domains. Generate detailed reports and analytics to provide insights into ad-blocking effectiveness and network usage
- 3.4.2 **Machine Learning-based Ad Detection:** Explore the use of machine learning algorithms to improve ad detection and blocking. Train models to recognize and block new forms of advertisements, including sponsored content and native ads.
- 3.4.3 **Customizable Blocking Rules:** Allow users to create custom blocking rules based on specific criteria, such as domain, URL, or content type. Provide a user-friendly interface for managing custom rules and whitelists.
- 3.4.4 **Integration with VPN Services:** Integrate the ad-blocking system with VPN services to provide enhanced privacy and security for users. Ensure consistent ad-blocking functionality across all devices, including mobile devices and IoT devices.
- 3.4.5 **Customizable Blocking Rules:** Allow users to create custom blocking rules based on specific criteria, such as domain, URL, or content type. Provide a user-friendly interface for managing custom rules and whitelists.
- 3.4.6 **Integration with Content Filtering Systems:** Integrate the ad-blocking system with content filtering systems to provide comprehensive protection against unwanted content, including ads, malware, and phishing attempts.
- 3.4.7 **Scalability and Performance Optimization:** Optimize system performance to handle large volumes of network traffic efficiently. Implement caching mechanisms and distributed processing to reduce latency and improve response times.

By incorporating these future enhancements, Next-Gen Adblocker System using Raspberry Pi can become even more accurate, efficient, and user-friendly, providing valuable insights and automation to educational institutions for effective attendance tracking and management.



APPENDIX

INSTALLATION

To install and set up a Next Gens Adblocker project from GitHub, follow these detailed steps:

1. **Install Python:** Ensure that Python is installed on your system. You can download and install the latest version of Python from the official website:

<https://www.python.org/downloads/>

2. **Install Git:** Install Git, You can download Git from the official website:

<https://git-scm.com/downloads>

3. **Install pihole and unbound DNS:** To install pi hole run the command:

`curl -sSL https://install.pi-hole.net | bash`
`sudo apt install unbound`

This will install them both, use the config files provided in the repository to set up them both in your network to initiate Ad Blocking.

4. **Clone the Repository:** Open your terminal or command prompt and navigate to the directory where you want to clone the project. Then run the following command to clone the repository:

`git clone https://github.com/yuneshwaran/AdBlocker`

5. **Create a Virtual Environment:** Navigate into the project directory using the following command:

`cd AdBlocker`

Run the following command to create a virtual environment named "venv":

`python3 -m venv venv`



- 6. Activate the Virtual Environment:** Activate the virtual environment using the appropriate command for your operating system:

source venv/bin/activate

- 7. Install Dependencies:** With the virtual environment activated, install the project dependencies using the following command:

pip install -r requirements.txt

This command will install all the necessary packages and libraries specified in the **requirements.txt** file.

- 8. Run the suricata rules fetch file:** To fetch the latest rules file run the command

./fetch_et_open.sh

This will download the latest Suricata5 rules from emerging threats website to your AdBlocker directory.

- 9. Create a Superuser (Optional):** This will take the downloaded rules as input and generate an updated ad list for your pi-hole.

python et2dns.py --rules <downloaded .rules file name>

This will create a folder /output if it doesn't exist and generates the ad list in that directory.

Now the Next-Gen Ad Blocking project components, including Pi-hole, Unbound DNS, and the Python ad list generation script, are successfully installed on the Raspberry Pi. The project is now ready for exploration, modifications, and deployment. To host the ad-blocking service, you can utilize the Raspberry Pi itself or explore other options such as local servers or dedicated hardware. With its low power consumption and compact size, the Raspberry Pi offers a convenient and cost-effective solution for hosting the Next-Gen Ad Blocking service within your local network.



CODING

Generate.py

```
1. import argparse
2. from idstools import rule
3. from datetime import date
4. import tldextract
5. from os import makedirs
6. from pprint import pprint
7. description = (
8.     "Given a file containing a list of Suricata rules, extract any blocked domains."
9. )
10. parser = argparse.ArgumentParser(description=description)
11. parser.add_argument(
12.     "--rules",
13.     required=True,
14.     help="The name of the file containing a list of Suricata rules, one rule per line.",
15. )
16. args = parser.parse_args()
17. rules_in_file = open(args.rules, "r")
18. categories = {
19.     "MALICIOUS": {
20.         "description": "Blocks malware, phishing, coin miners, PUPs, exploits, etc.",
21.         "utility": "High - useful at home and in corporate environments",
22.         "count": 0,
23.         "tags": [
24.             "ET TROJAN",
25.             "ET MALWARE",
26.             "ET MOBILE_MALWARE",
27.             "ET CURRENT_EVENTS",
28.             "ET PHISHING",
29.             "ET ATTACK_RESPONSE",
30.             "ET ADWARE_PUP",
31.             "ET EXPLOIT_KIT",
32.             "ET WEB_CLIENT",
33.             "ET WEB_SERVER",
34.             "ET COINMINER",
35.         ],
36.     },
37.     "SUSPICIOUS": {
38.         "description": "Blocks link shorteners, pastebin services, games, etc.",
39.         "utility": "Moderate - useful in strict corporate environments, maybe not at home",
40.         "count": 0,
```



```
40.     "tags": [  
41.         "ET INFO DYNAMIC_DNS",  
42.         "ET POLICY",  
43.         "ET GAMES",  
44.         "ET DNS",  
45.     ], },  
46.     "INFORMATIONAL": {  
47.         "utility": "Low - may be useful in certain strict corporate environments",  
48.         "count": 0,  
49.         "tags": ["ET INFO", "ET HUNTING"],},},  
50. makedirs("output/", exist_ok=True)  
51. malicious_file = open("output/malicious.txt", "w")  
52. suspicious_file = open("output/suspicious.txt", "w")  
53. informational_file = open("output/informational.txt", "w")  
54. def write_by_category(category, write_this):  
55.     if category == "MALICIOUS":  
56.         malicious_file.write(f"{write_this}\n")  
57.     if category == "SUSPICIOUS":  
58.         suspicious_file.write(f"{write_this}\n")  
59.     if category == "INFORMATIONAL":  
60.         informational_file.write(f"{write_this}\n")  
61. header = """"# (Unofficial) Emerging Threats PiHole blocklist  
62. # https://github.com/tweedge/emerging-threats-pihole  
63. # Category: {}  
64. # Description: {}  
65. # Utility: {}  
66. # Status: Beta / in development  
67. # Last modified: {}  
68. # Blocklist generated"""  
69. for category_name, category_content in categories.items():  
70.     write_by_category(  
71.         category_name,  
72.         header.format(  
73.             category_name,  
74.             category_content["description"],  
75.             category_content["utility"],  
76.             date.today(), ),)  
77. for line in rules_in_file:  
78.     input_rule = line.strip()  
79.     if input_rule == "":  
80.         Continue  
81.     if input_rule[0] == "#":  
82.         Continue
```



```
83.  parsed_rule = rule.parse(input_rule)
84.  if not "dns.query" in parsed_rule.keys() and not "dns_query" in parsed_rule.keys():
85.      continue
86.  may_be_component_of_domain = True
87.  if "endswith" in parsed_rule.keys():
88.      may_be_component_of_domain = False
89.  if "isdataat" in parsed_rule.keys():
90.      if parsed_rule["isdataat"] == "!1,relative":
91.          may_be_component_of_domain = False
92.  if may_be_component_of_domain:
93.      continue
94.  if "pcr" in parsed_rule.keys():
95.      if option["name"] == "content":
96.          if option["value"].startswith("!"):
97.              mixed_allow_and_deny = True
98.  if mixed_allow_and_deny:
99.      clean_domain = parsed_rule.content.strip('.')
100.         parsed_domain = tldextract.extract(clean_domain)
101.         # skip any suspicious TLDs as those are unsuitable for DNSBL
102.         if parsed_domain.domain == "" or parsed_domain.suffix == "":
103.             continue
104.         message = parsed_rule.msg
105.         category = ""
106.         for category_name, category_content in categories.items():
107.             for category_tag in category_content["tags"]:
108.                 if category_tag in message and not category:
109.                     category = category_name
110.                     categories[category_name]["count"] += 1
111.         if not category:
112.             print(f'Couldn't categorize message: {message}')
113.             write_by_category(category, f'0.0.0.0\t{clean_domain}')
114.         if categories["MALICIOUS"]["count"] < 1500:
115.             print("There are too few MALICIOUS domains -- is the input file correct?")
116.             exit(1)
```



SUMMARY OF FINDINGS

1. Effective Ad Blocking:

- The network-wide ad blocker, implemented using Pi-hole, Unbound DNS, and Python ad list generation with Suricata rules, has proven to be highly effective in blocking ads across the entire network.
- By intercepting DNS requests and filtering out ad-serving domains, the system ensures that users experience an ad-free browsing environment.

2. Improved network Security:

- In addition to ad blocking, the system enhances network security by preventing access to known malicious domains and tracking scripts.
- By blocking these threats at the DNS level, the system provides an additional layer of defense against malware, phishing attempts, and other online threats.

3. Ease of Use and Deployment:

- The integration of Pi-hole and Unbound DNS on Raspberry Pi provides a plug-and-play solution for ad blocking without the need for complex installation steps.
- Users can easily deploy the ad blocker on their local network by following simple setup instructions, making it accessible to both novice and experienced users.

4. Dynamic Ad list Generation:

- The Python script developed for dynamic ad list generation using Suricata rules ensures that the ad-blocking lists are continuously updated with the latest threats.

5. Scalability and Performance:

- The system is highly scalable and capable of handling network traffic efficiently, ensuring minimal impact on network performance.
- Even in environments with high volumes of network traffic, the ad blocker maintains its effectiveness without compromising network speed or performance.



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